ICT261 · SKYRIDGE LODGE BILLING SYSTEM

Room Availability Management

Member 7 – Implementation Summary & Handover Documentation

Member
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Course
ICT261 – Object-Oriented Programming

GitHub repository
https://github.com/GamerVal022/ICT261

Working branch
feature/member-7-room-availability

1. Overview

This document describes the Room Availability Management functionality implemented by Member 7 in the SkyRidge Lodge Billing System.

The objective was to introduce room availability tracking so administrators and customers can view the availability of different room categories. The implementation extends the existing lodge management and reservation functionality with availability summaries, room counts, percentages and visual indicators that respond to reservation activity.

The four room categories represented are Standard Room, Deluxe Room, Family Room and Thatched Deluxe Cottage.

2. Administrator Room Availability

The administrator interface includes a Room Availability Overview showing a summary of the lodge's room inventory. The overview distinguishes available, occupied, reserved and maintenance rooms.

- Total number of rooms.
- Available rooms.
- Occupied rooms.
- Reserved rooms.
- Rooms under maintenance.
- Occupancy information.

The room inventory also supports filtering by room type and room status, helping administrators locate rooms.

Availability states

Status	Meaning
Available	Sufficient rooms remain available.
Limited	Fewer than 30% of the rooms remain available.
Fully Booked	No rooms remain available.

3. Customer Room Availability

The customer booking interface includes a Room Availability Overview. Customers can view availability for the four room categories before making a reservation.

Each summary card displays the room category, available rooms, total rooms, availability information and a visual progress indicator.

Progress indicators and percentages

The progress bar represents the proportion of rooms available. The accompanying unavailable percentage represents the proportion of rooms that are not available.

Available rooms	Progress bar	Unavailable percentage
3 of 3	100% available	0%
2 of 3	Approximately 67% available	Approximately 33%
1 of 3	Approximately 33% available	Approximately 67%
0 of 3	0% available	100%

The interface uses visual status indicators, including green for available and red for fully booked.

4. Reservation and Availability Updates

The availability implementation works alongside the existing customer reservation functionality. Customers use the existing reservation interface to select check-in and check-out dates, choose a room category and make a reservation.

After a reservation is made, the availability overview updates to reflect the reduced number of available rooms. For example, a category with four available cottages can display the following progression:

Reservation stage	Available cottages
Before reservations	4 of 4
After first reservation	3 of 4
After second reservation	2 of 4
After third reservation	1 of 4
After fourth reservation	0 of 4

> **Scope clarification:** The reservation form and its wider booking workflow are existing project functionality. Member 7's contribution focuses on room availability tracking, its presentation and its integration with that workflow.

5. Availability Rules

1. Availability is presented separately for each room category.
2. Available rooms are displayed against the total room count.
3. The availability progress indicator represents the available proportion.
4. The unavailable percentage represents the proportion of rooms that are not available.
5. A category with no available rooms is marked Fully Booked.
6. A category with fewer than 30% of its rooms available is marked Limited.
7. Rooms under maintenance are not offered for customer reservations.
8. Reservation activity is reflected in the customer availability overview.

The administrator inventory overview separately identifies available, occupied, reserved and maintenance room statuses.

6. Room Categories and Rates

Room category	Rate per night	Capacity
Standard Room	K350	2 persons
Deluxe Room	K550	3 persons
Family Room	K750	5 persons
Thatched Deluxe Cottage	K1,200	4 persons

These are the room type values configured in the Member 7 implementation.

7. Main Files Involved

File	Responsibility
`RoomAvailability.java`	Represents room availability information and availability calculations.
`RoomAvailabilityService.java`	Retrieves and calculates room availability information.
`RoomType.java`	Defines room categories, rates and capacities.
`RoomsController.java`	Handles the administrator room interface and availability overview.
`CustomerController.java`	Handles the customer availability overview and its interaction with the booking interface.
`rooms.fxml`	Defines the administrator room interface.
`customer.fxml`	Defines the customer booking and availability interface.

The implementation works with existing project services, including `RoomService`, `BookingService` and `DataStore`.

8. Testing and Verification

The following checks were completed during development and integration:

Test	Result
Application build	**Passed**
Automated project tests	**Passed**
Administrator availability overview	**Passed**
Room status filtering	**Passed**
Updating room status and availability totals	**Passed**
Customer availability overview	**Passed**
Availability progress indicators and percentages	**Passed**
Reservation updates reflected in availability	**Passed**
Fully booked category restriction	**Passed**
Check-in/check-out date selection checks	**Passed**
Administrator/customer availability consistency checks	**Passed**

Commands used

Run automated tests:

```
.\gradlew.bat clean test
```

Run the application:

```
.\gradlew.bat run
```

The project was tested after integrating the updated upstream changes.

9. Handover Notes

1. Preserve the existing room availability model and service when modifying the room management module.
2. Check availability-related functionality before replacing or substantially changing `CustomerController`.
3. Keep administrator and customer availability displays consistent with the underlying room and booking data.
4. Retest reservations after changes to booking or room management logic.
5. Ensure rooms under maintenance are not offered for customer reservations.
6. When modifying date-based availability, test overlapping reservations across different check-in and check-out dates.
7. Check stored-data compatibility before changing room category identifiers.

10. Conclusion

Member 7 implemented Room Availability Management as an extension of the SkyRidge Lodge Billing System. The work provides administrators with an overview of room inventory and availability and allows customers to view room availability through the customer booking interface.

The availability indicators, room counts and percentages respond to reservation activity. The implementation was integrated with the updated upstream project and underwent build, automated and manual functional testing.